Statistics - Histogram estimators

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Chapter 1

Introduction

1.1 Notations and context

Let us take $(X_1, ..., X_n)$ an $n-$sample in $E = [0, 1]$ with a density p with respect to the Lebesgue measure λ. We will work in the case of a regular partition $\mathcal{I}$ of E into D equal sub-intervals.

We would like to estimate p, and in the course we have studied histogram estimators; The histogram of p based on the partition $\mathcal{I}$ is the random function:

$$\hat{p}_{\mathcal{I}}(x) = \sum_{I \in \mathcal{I}} \frac{N_I}{n\lambda(I)} \mathbb{1}_I(x)$$

where N_I is 'number' of data which fall into I, more precisely: $N_I = \sum_{i=1}^{n} \mathbb{1}_{X_i \in I}$. Since we have chosen a regular partition, we have in particular, for all $I \in \mathcal{I}, \lambda(I) = \frac{1}{D}$.

Finally, let us denote by $V_{\mathcal{I}}$ the linear space of step functions: $\sum_{I \in \mathcal{I}} a_I \mathbb{1}_I$ where $a_I \in \mathbb{R}$. We recall the L^1-risk of the histogram as:

$$R_1 = \mathbb{E}_p(||p - p_{\mathcal{I}}||_1)$$

Moreover, we have the following bound (for regular partitions into D intervals):

$$\mathbb{E}_p(||p - p_{\mathcal{I}}||_1) \leq 2 \inf_{f \in V_{\mathcal{I}}} ||p - f||_1 + \sqrt{\frac{D-1}{n}}$$

Suppose we would like to find numerically a histogram $\bar{p}$ in $V_{\mathcal{I}}$ such that $||p - \bar{p}_D||_1$ is as small as possible. The classical approach (in case we know p) would be to compute for each choice of partition $\mathcal{I}$ (depending on D), the quantity $\int_0^1 |p(x) - \hat{p}_D(x)|dx$ and therefore to deduce the value of D which achieves this minimum.

Since there has to be a tradeoff between the bias and the variance, we shall not choose a D which is too large.

In this paper we will attempt to compare the above procedure with another procedure, suggested by Birge and Rozenholc, which would allow us to select a suitable value $\hat{D}$ from the data.

We already know a rule of thumb to make this choice, for instance the Sturge's rule which suggests to take a number of bins close to $log_2(50) + 1$. However this rule is very general and depends only on the sample size and not on the particular distribution.

Let us describe below the new suggested procedure and how we will implement it.

1.2 Procedure description

For all $D \geq 1$, we define the log-likelihood at $\hat{p}_D$ based on the observations $\mathbb{X} = (X_1, ...X_n)$ by:

$$L_{\mathbb{X}}(D) = \sum_{I \in I_D} N_I log(N_I \frac{D}{n})$$

The procedure suggests to consider a penalty function, defined for all integer $D > 0$ as: $pen(D) = D - 1 + (log(D))^{2.5}$. The method suggests that the value of $\hat{D}$ to choose is:

$$\boxed{\hat{D} = \underset{1 \leq D \leq n/log(n)}{argmax} \left(\sum_{I \in \mathcal{I}_D} N_I log(N_I D/n) - (D - 1 + (log(D)^{2.5}) \right)}$$

Therefore, for any given density p, we compute the $\hat{D}$ above and we will compare it with the value of D which achieves $M = \underset{1 \leq D \leq n/log(n)}{min} \int_0^1 |p(x) - \hat{p}_D(x)| dx$. This is made possible by computing the ratio:

$$R = \frac{\int_0^1 |p(x) - \hat{p}_{\hat{D}}(x)| dx}{M}$$

A ratio of 1 suggests that $\hat{D} = D_{min}$ (the two 'D' values coincide).

We will repeat the experience various times and then compute the mean value and variance of the ratios, which we denote by:

$$\bar{R} = \frac{1}{k} \sum_{j=1}^{k} R_j \qquad \bar{V} = \frac{1}{k} \sum_{j=1}^{k} (R_j - \bar{R})^2$$

.

We will give below the explicit expressions of the densities we have used (we used the 3 densities suggested but we also tried 2 other densities of our choice).

1.3 Generating an n-sample and choice of densities

To generate an n-sample with density p we will use the fundamental theorem of simulation. Let us denote by U a random variable with uniform distribution on $[0;1]$.

Theorem 1: Simulation of an n-sample

Let X be a real-valued random variable with cumulative distribution function $F(x) = \mathbb{P}(X \leq x)$, we denote by F^{-1} its generalized inverse defined as $F^{-1}(u) = inf\{x : F(x) \geq u\}$. We know that $F^{-1}(U)$ has the same distribution as X.

Therefore, in order to generate an n-sample, it is sufficient to sample from a uniform random distribution and to invert the cumulative distribution function of each of the distributions we consider.

Let us determine each cumulative distribution function F_k and its inverse F_k^{-1} for each density p_k:

$p_1 = \frac{1}{2}\mathbb{1}_{[0,1/2)} + \frac{3}{2}\mathbb{1}_{[1/2;1]}$

$p_2 = 2\mathbb{1}_{[0,1/8]} + \frac{4}{5}\mathbb{1}_{[1/8;3/4)} + \mathbb{1}_{[3/4;1]}$

$p_3 = \frac{1}{Arctan(1)(1+x^2)}\mathbb{1}_{[0,1]}$

We have, respectively:

$F_1(x) = \frac{x}{2}\mathbb{1}_{[0,1/2])} + (\frac{3x}{2} - \frac{1}{2})\mathbb{1}_{[1/2,1]}$

$F_2(x) = 2x\mathbb{1}_{[0,1/8)} + (\frac{4x}{5} + \frac{15}{100})\mathbb{1}_{[1/8,3/4)} + x\mathbb{1}_{[3/4,1]}$

$F_3(x) = \frac{Arctan(x)}{Arctan(1)}\mathbb{1}_{[0,1]}$

Thus: $F_1^{-1}(u) = 2u\mathbb{1}_{[0,1/4]} + (\frac{2u}{3} + \frac{1}{3})\mathbb{1}_{[1/4,1]}$

$F_2^{-1}(u) = \frac{u}{2}\mathbb{1}_{[0,1/4)} + (\frac{5u}{4} - \frac{3}{16})\mathbb{1}_{[1/4,3/4]} + u\mathbb{1}_{(3/4,1]}$

$F_3^{-1}(u) = tan(\frac{\pi}{4}u)\mathbb{1}_{[0,1]}$

We have also tried the two following densities (for which we will show the results we have obtained):

$p_4(x) = 2(3x - floor(3x))\mathbb{1}_{[0,1]}$

$p_5(x) = \frac{\pi}{3}sin(2\pi x)\mathbb{1}_{[0,1/2]} + \frac{2\pi}{3}sin(2\pi(x - \frac{1}{2}))\mathbb{1}_{[1/2,1]}$.

Other possible interesting choices of underlying densities would be functions which "vary a lot" (high total variation) since by the course, we recall that:

$$\inf_{f \in V_{\mathcal{I}}} ||p - f||_1 \leq \frac{1}{2D} V_{[0,1]}(p)$$

Chapter 2

Numerical Implementation 2026 update - Reworked on GitHub

Below we explain the main steps of our implementation in Python. We explain what each command does, at least for the first density p_1. Please find in the appendix the full version of the code (which is not present in this section, some steps have been omitted for readability).

The first table that appears below is the distribution of our n-sample. We also implement a simple integration by Riemann sums below.

We would also like to state that we chose to take the final D_M (the value which achieves the optimal distance) as the modal value in the set of all the D's which achieve this optimal distance (since we have repeated the simulation several times). The same applies for $\hat{D}$.

Chapter 3

Results and conclusion

3.1 Interpretation

Let us present below the results we have obtained for our simulations. We have attempted several sample sizes $n = 50, 100, 200$.

Since we wanted to see if the values of R were centered around 1 for each density, we computed the relative standard deviation of the statistical observations of R. The relative standard deviation indicates how dispersed the values of R we obtained are.

$$RSD = \frac{\sqrt{\bar{V}}}{\bar{R}}$$

.

We interpret a high RSD as a high dispersion around the mean value $\bar{R}$: in this case, we may not recommend the procedure. This is indeed the case for the density p_1 for instance.

For $n = 50$, we notice that the RSD is highest for the first and third densities, and lower for the second one. While the density p_3 has a relatively high RSD at first, we can clearly see that $\bar{R}$ is quite large (around 2) - which also seems to apply for our density p_5.

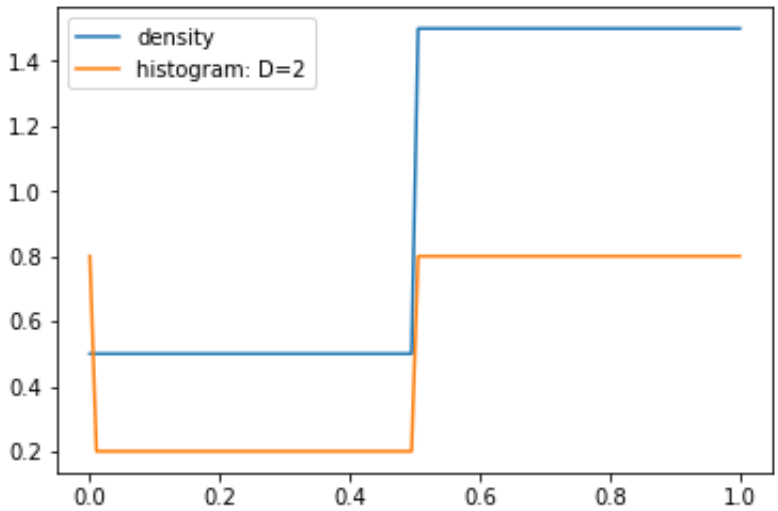
	n=50					n=100					n=200				
density	D_M	D_hat	RSD	R_bar	V_bar	D_M	D_hat	RSD	R_bar	V_bar	D_M	D_hat	RSD	R_bar	V_bar
p1	2.0	2.0	0.636129	1.152265	0.537273	2.0	2.0	0.194199	1.029857	0.039999	2.0	2.0	0.058995	1.010767	0.003556
p2	8.0	1.0	0.285209	1.527161	0.189713	8.0	1.0	0.235900	1.582066	0.139286	8.0	8.0	0.246560	1.532948	0.142856
p3	2.0	1.0	0.617571	1.977495	1.491434	2.0	1.0	0.543543	2.502888	1.850765	2.0	8.0	0.372258	3.199492	1.418566
p4	2.0	3.0	0.154479	1.347643	0.043340	2.0	6.0	0.123503	1.276346	0.024848	2.0	6.0	0.080536	1.185111	0.009110
p5	9.0	2.0	0.172119	1.440106	0.061439	11.0	7.0	0.162123	1.389916	0.050777	12.0	7.0	0.172859	1.366173	0.055769

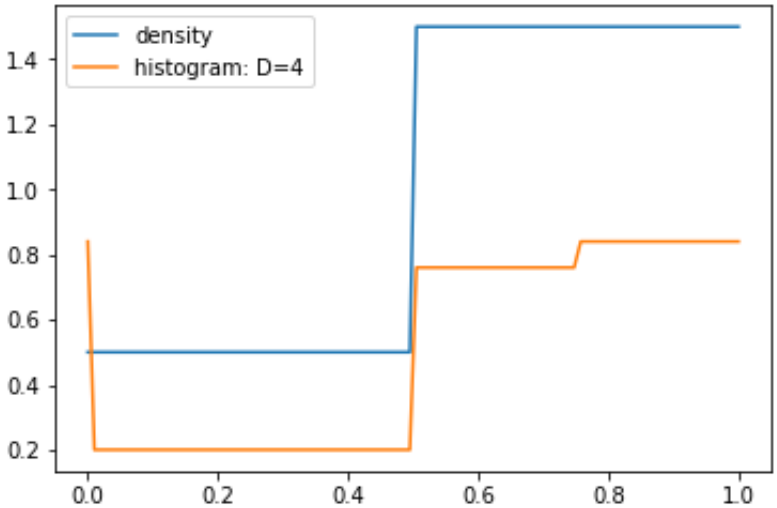
An interesting pattern we observed was the effect of taking a larger n: we can see that the RSD (resp. the variance) decreases in general, but the $\bar{R}$ for the density number 3 is always the largest.

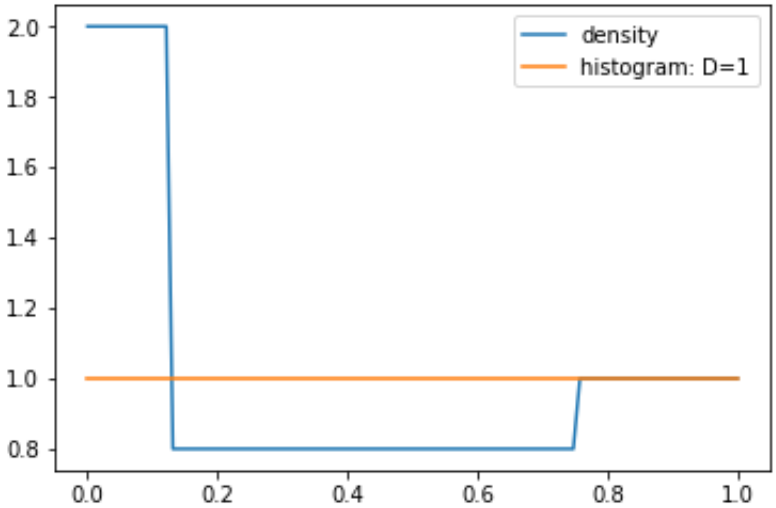
Finally, we can also observe that the optimal D (D_M) is not always close to Sturge's rule (density p_2 for $n = 50$, and density p_5).

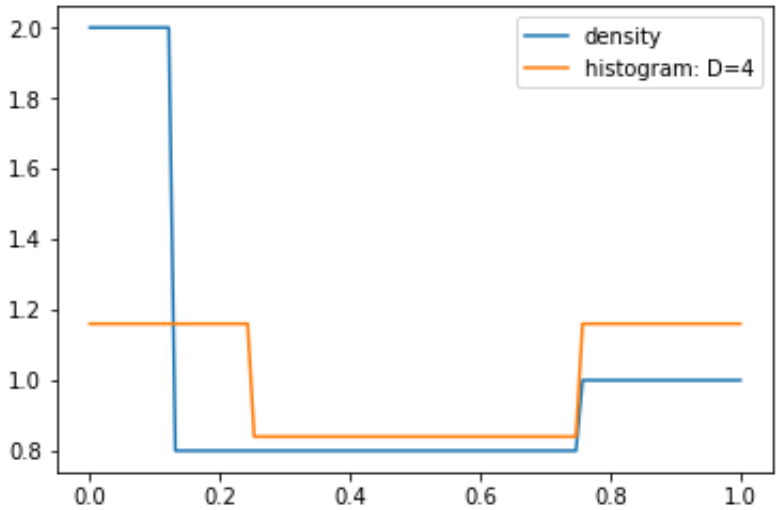
3.2 Histogram graphs

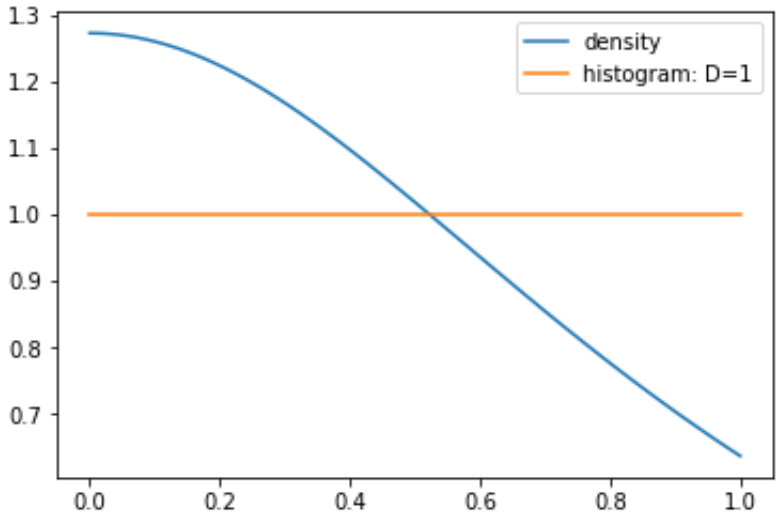
Please find below some graphs which show the densities p_1 to p_5 in order, and the two histograms we retained for each (the one for D_M and the one for $\hat{D}$), as indicated in the table above.

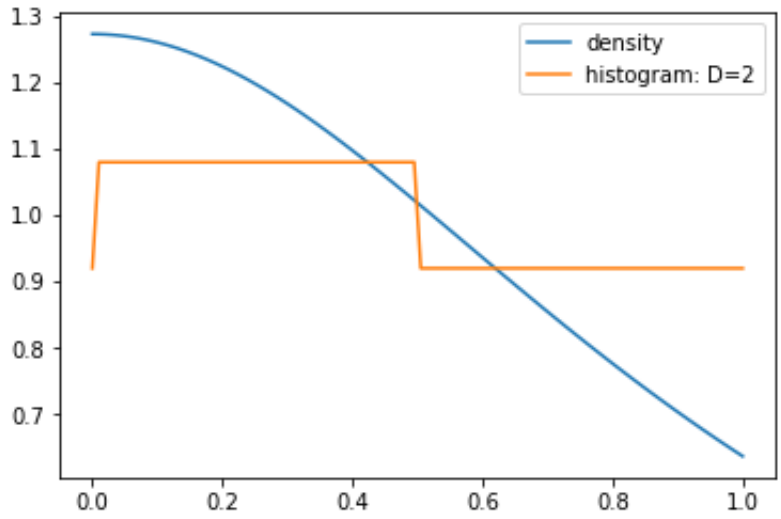

density
histogram: D=2
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0

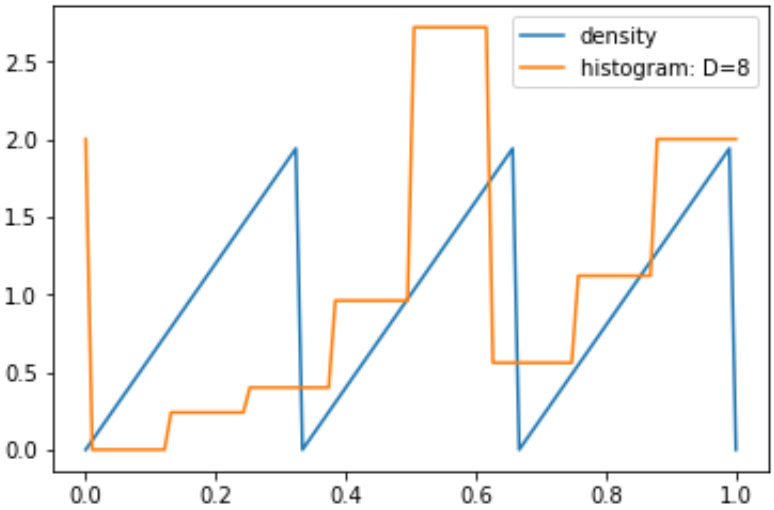

density
histogram: D=4
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0

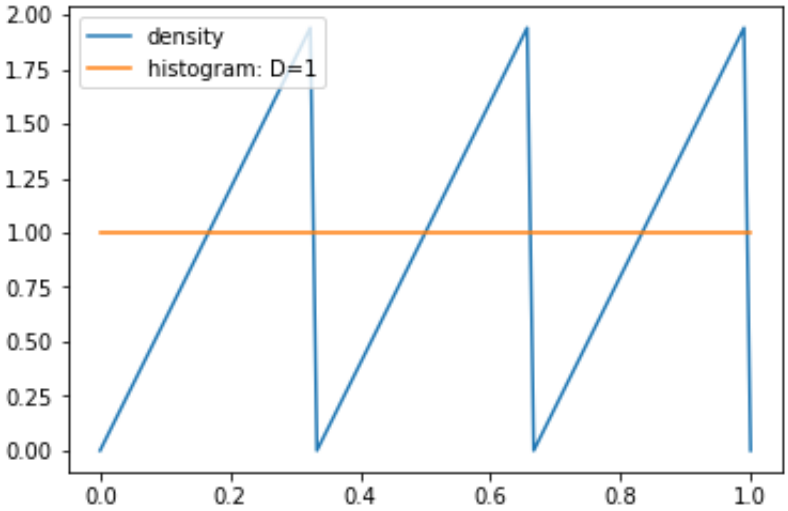

density
histogram: D=1
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.0
0.2
0.4
0.6
0.8
1.0

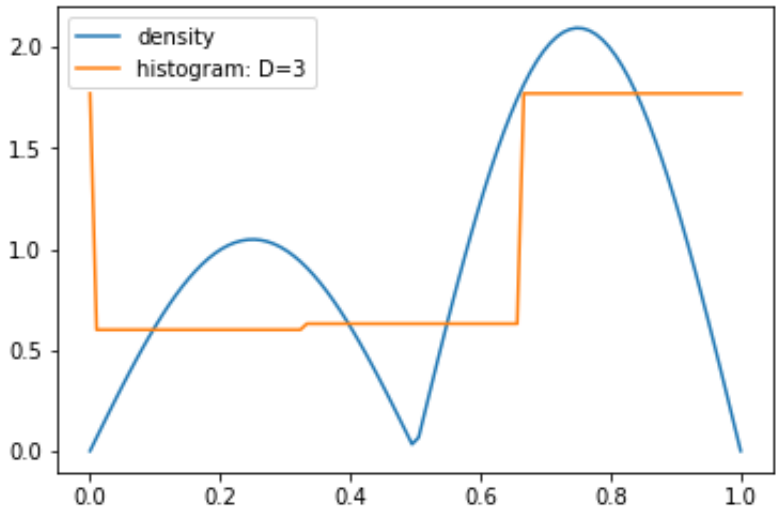

density
histogram: D=4
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.0
0.2
0.4
0.6
0.8
1.0

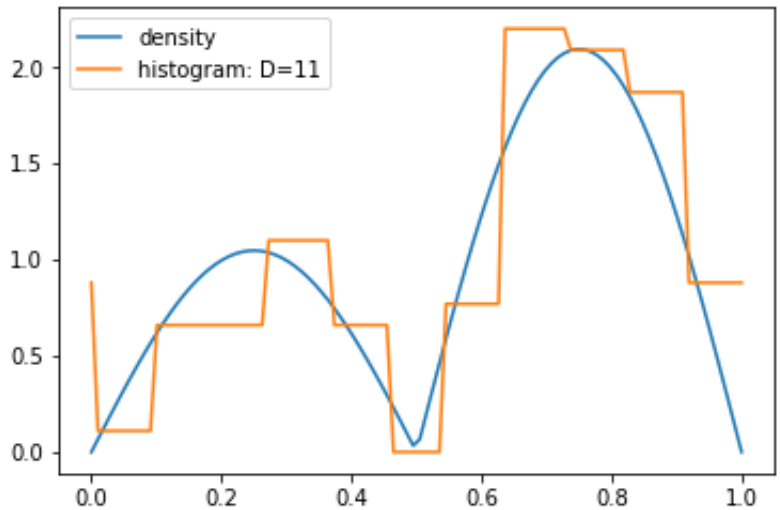

density
histogram: D=1
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.0
0.2
0.4
0.6
0.8
1.0


density
histogram: D=2
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.0
0.2
0.4
0.6
0.8
1.0


density
histogram: D=8
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0


density
histogram: D=1
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0


density
histogram: D=3
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0


density
histogram: D=11
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0